

Systems Biology: Mathematics for Biologists

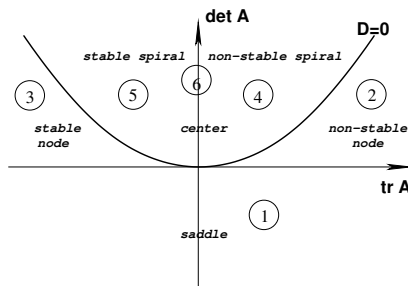


Levien van Zon, Theoretical Biology, UU

Last Time: Plan of Attack

Given a general 2D ODE
(Ordinary Differential Equation)
system:

$$\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$$



- 1 Write $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$, draw the null-clines, draw the vectorfield, find equilibria.
- 2 Look at vectorfield to determine equilibrium types.
- 3 If 2 fails: Use a graphical jacobian to determine the equilibrium type (using the determinant-trace method).
- 4 If 3 fails: Use an analytical jacobian to determine the equilibrium type (using the determinant-trace or the eigenvalue method).

Equilibrium types so far

Eigenvalues of Jacobian J :

$$\lambda_{1,2} = \frac{\text{tr } J \pm \sqrt{D}}{2}$$

$$J = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}$$

$$\det J = ad - bc$$

$$\text{tr } J = a + d$$

Equilibrium types so far

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$$\lambda_{1,2} = \frac{\text{tr } J \pm \sqrt{D}}{2}, \text{ with}$$

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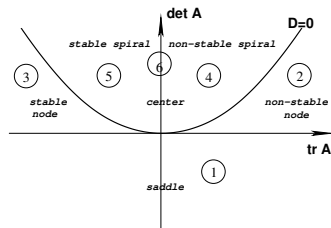
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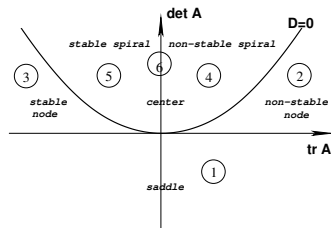
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- saddle node: $\det < 0$, real λ s, one positive, one negative



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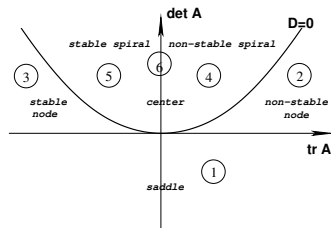
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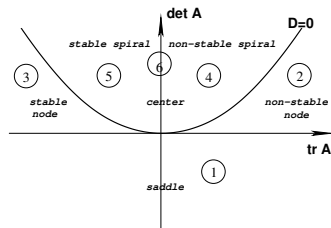
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- **Unstable equilibria:**

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- unstable node: $\text{tr} > 0$, real, positive eigenvalues

- **Stable equilibria:**

- stable node: $\text{tr} < 0$, real, negative eigenvalues



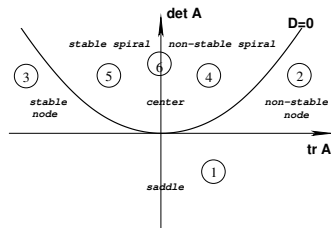
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- **Unstable equilibria:**

- saddle node: $\det < 0$, real λ s, one positive, one negative
- unstable node: $\text{tr} > 0$, real, positive eigenvalues
- unstable spiral: $D < 0$, complex λ s, positive real part

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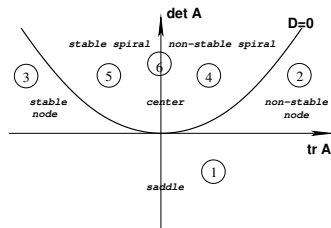
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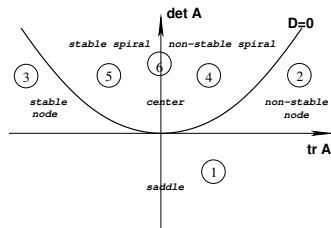
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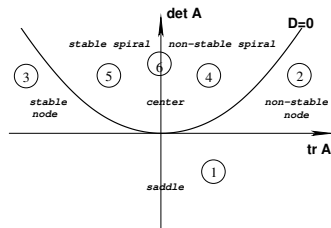
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Note that these equilibria are all **single points**, so we can find a Jacobian.

Limit cycles

The “classical” Lotka-Volterra model

Consider again the Lotka-Volterra predator-prey model.

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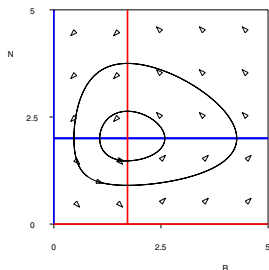
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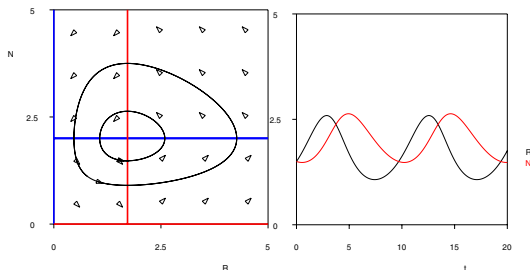


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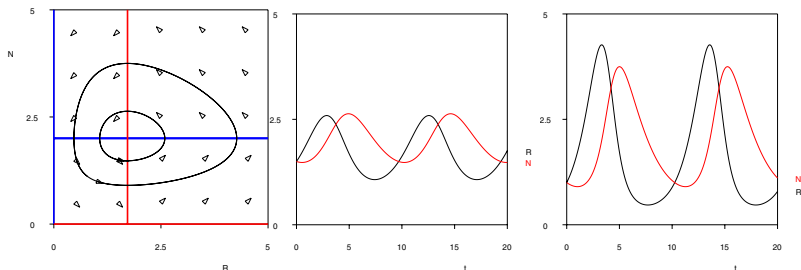


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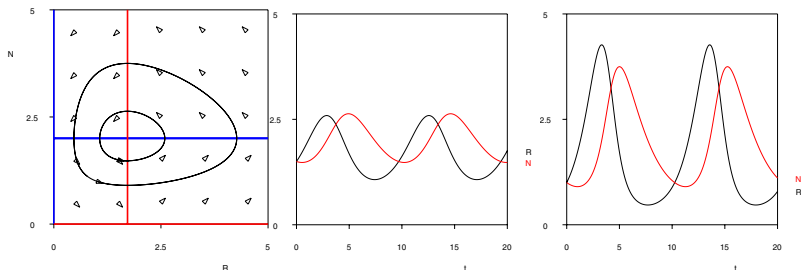


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Oscillations around a center point. But this depends on rather unrealistic null-clines!

Parameter change in the LV-model

Consider a slightly more realistic Lotka-Volterra predator-prey model:

$$\begin{cases} \frac{dR}{dt} = rR(1 - \frac{R}{K}) - bNF \\ \frac{dN}{dt} = bNF - dN \end{cases} \quad \text{with} \quad F = \frac{R}{h + R}$$

and with $b = 0.5$, $d = 0.43$, $h = 0.1$, $r = 1$, variable K .

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and with $b = 0.5$, $d = 0.43$, $h = 0.1$, $r = 1$, variable K .

We will keep all parameters fixed except for K .

We study what happens with this system if we change K .

Parameter change in the LV-model

$$\begin{cases} \frac{dR}{dt} = rR\left(1 - \frac{R}{K}\right) - bNR\frac{1}{h+R} \\ \frac{dN}{dt} = bNR\frac{1}{h+R} - dN \end{cases} \quad \text{with} \quad b = 0.5, d = 0.43, h = 0.1, r = 1$$

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$\frac{dR}{dt} = 0$ gives two R -nullclines:

$$R = 0 \quad \text{and} \quad r(1 - \frac{R}{K}) - bN\frac{1}{h+R} = 0$$

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$\frac{dN}{dt} = 0$ gives two N -nullclines:

$$N = 0 \quad \text{and} \quad R = \frac{dh}{b-d} \approx 0.61$$

Parameter change in LV-model

Next, we determine the equilibria:

$$\begin{cases} \frac{dR}{dt} = rR(1 - \frac{R}{K}) - bNR\frac{1}{h+R} = 0 \\ \frac{dN}{dt} = bNR\frac{1}{h+R} - dN = 0 \end{cases}$$

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Start with the easiest equation, here the bottom one.

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This gives us $rR(1 - \frac{R}{K}) = 0$, or $rR - \frac{r}{K}R^2 = 0$,

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$$\text{or } N = \frac{2b(bK - d(h+K))r(\frac{dh}{b-d})}{(b-d)dK}$$

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This gives us

$$r\frac{dh}{b-d}\left(1 - \frac{\frac{dh}{b-d}}{K}\right) - bN\frac{\frac{dh}{b-d}}{h + \frac{dh}{b-d}} \approx 0.61\left(1 - \frac{0.61}{K}\right) - (0.5 \times 0.61)\frac{N}{0.1+0.61} = 0$$

$$\text{or } N = \frac{2b(bK - d(h+K))r(\frac{dh}{b-d})}{(b-d)dK} \approx 1.43 - 0.87\frac{1}{K}$$

Parameter change in LV-model

Next, we determine the equilibria:

$$\begin{cases} \frac{dR}{dt} = rR(1 - \frac{R}{K}) - bNR\frac{1}{h+R} = 0 \\ \frac{dN}{dt} = bNR\frac{1}{h+R} - dN = 0 \end{cases} \quad \text{with } b = 0.5, d = 0.43, h = 0.1, r = 1$$

Start with the easiest equation, here the bottom one.

$\frac{dN}{dt} = 0$ gave us two solutions: $N = 0$ and $R = \frac{dh}{b-d} \approx 0.61$

First, substitute $N = 0$ into the top equation $\frac{dR}{dt} = 0$:

This gives us $rR(1 - \frac{R}{K}) = 0$, or $rR - \frac{r}{K}R^2 = 0$,

with solutions $R = 0 \vee R = K$.

So the first two equilibria are $(0, 0)$ and $(K, 0)$.

Next, substitute $R = \frac{dh}{b-d} \approx 0.61$ into the top equation $\frac{dR}{dt} = 0$:

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$$r\frac{dh}{b-d}(1 - \frac{\frac{dh}{b-d}}{K}) - bN\frac{\frac{dh}{b-d}}{h + \frac{dh}{b-d}} \approx 0.61(1 - \frac{0.61}{K}) - (0.5 \times 0.61)\frac{N}{0.1+0.61} = 0$$

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So the third equilibrium is $(0.61, 1.43 - 0.87\frac{1}{K})$.

Parameter change in LV-model

What happens with the non-trivial equilibrium if we change K ?

Parameter change in LV-model

What happens with the non-trivial equilibrium if we change K ?

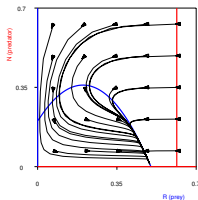
That depends on where the $R = 0.61$ predator nullcline intersects the parabola of the prey-nullcline (which has its top at $R = \frac{K-h}{2}$):

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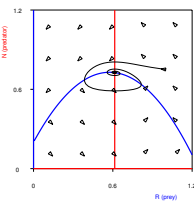
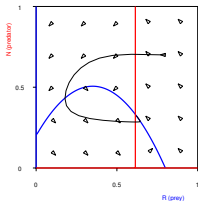
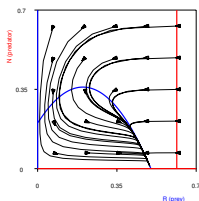


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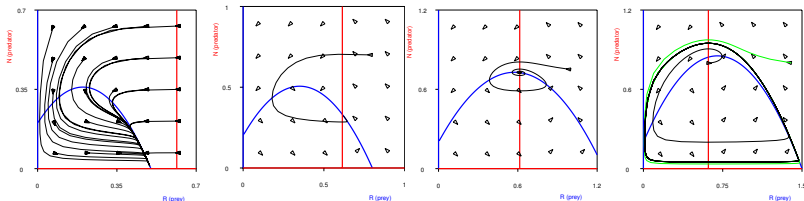


Parameter change in LV-model

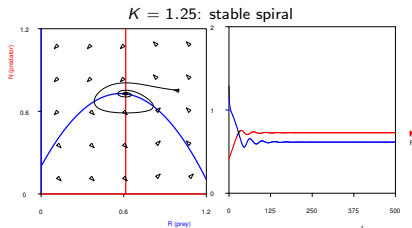
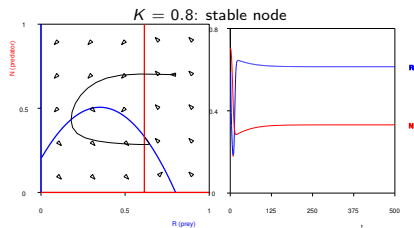
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- If $K > 1.33$, it intersects to the left of the top (as $\frac{K-0.1}{2} > 0.61$). We have an unstable spiral.

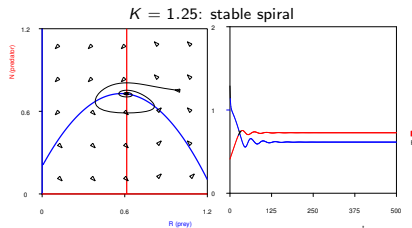
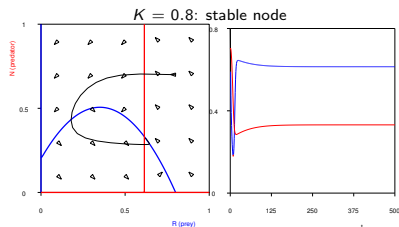


Parameter change in LV-model

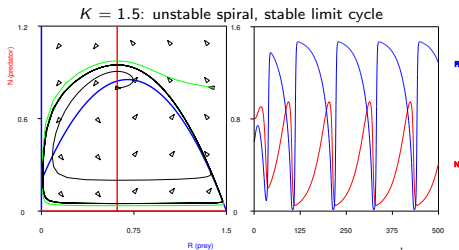


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Parameter change in LV-model



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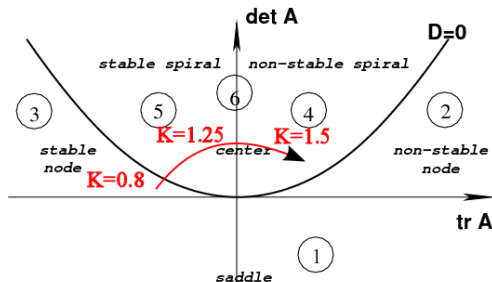


The **global** vectorfield remains the same, the **local** vectorfield is now qualitatively different. It has changed from stable to unstable.

Parameter change in LV-model

So in the determinant-trace scheme:

Increasing K results in series of changes in dynamics:

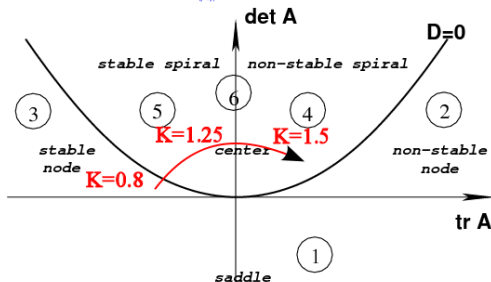
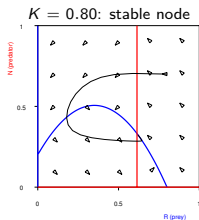


Parameter change in LV-model

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stable node



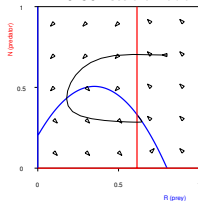
Parameter change in LV-model

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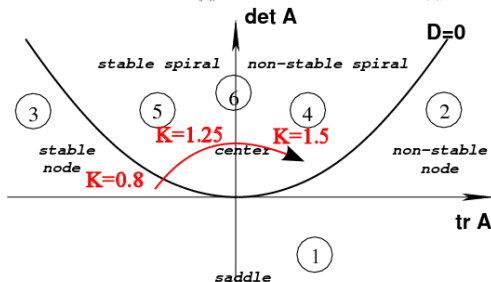
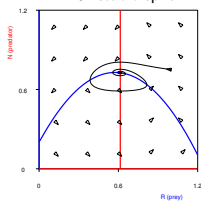
Increasing K results in series of changes in dynamics:

stable node $\rightarrow$ stable spiral

$K = 0.80$: stable node



$K = 1.25$: stable spiral



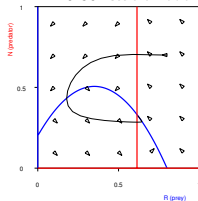
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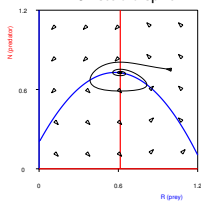
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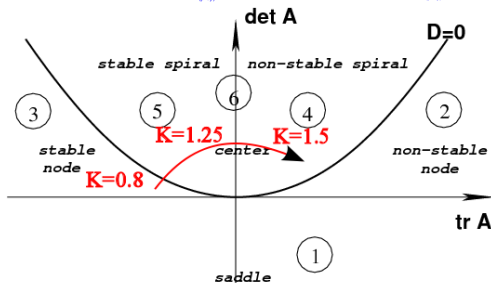
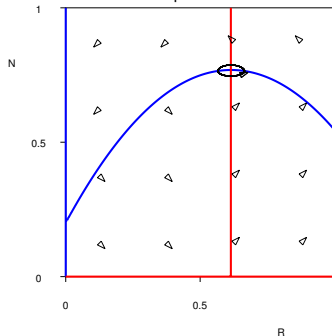
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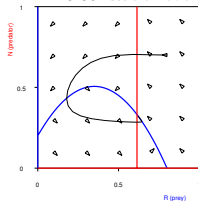
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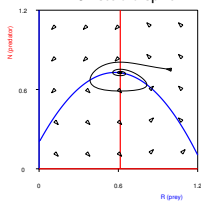
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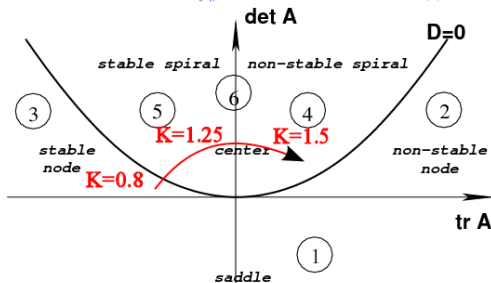
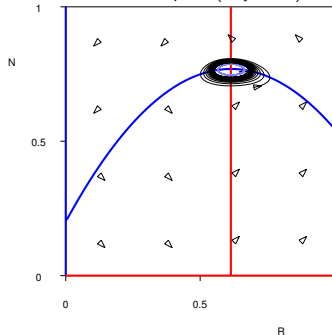
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$K = 1.25$: stable spiral



$K = 1.33$: center point (only local!)



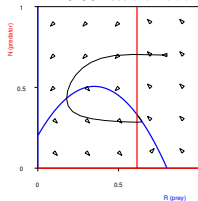
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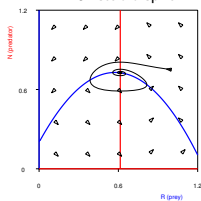
Increasing K results in series of changes in dynamics:

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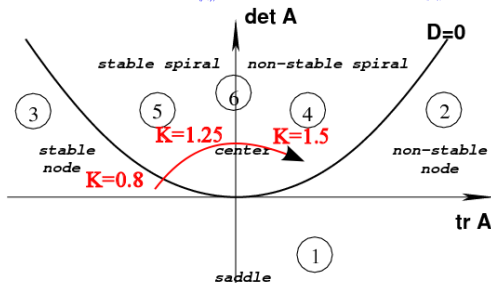
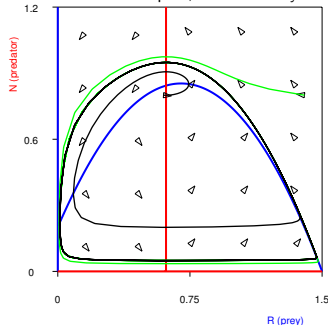
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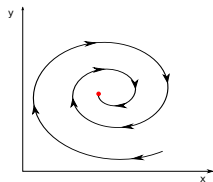


$K = 1.5$: unstable spiral, stable limit cycle



Point and Cycle attractors

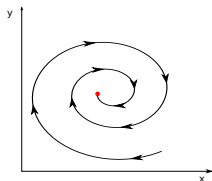
A stable equilibrium is a *point attractor*



Point and Cycle attractors

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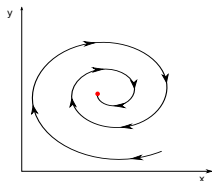
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Point and Cycle attractors

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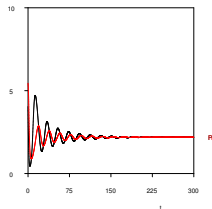
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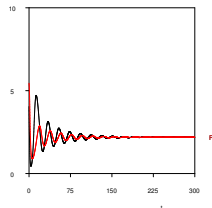
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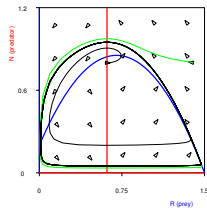
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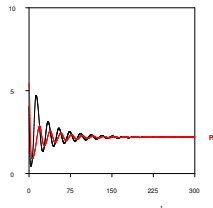
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Point and Cycle attractors

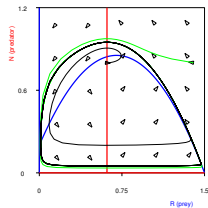
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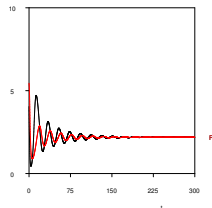
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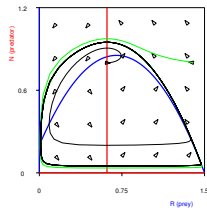
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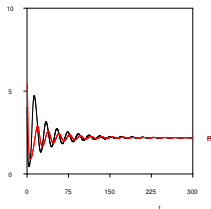
- It's a series of (x, y) points, forming closed curve.
- Once it is reached, the system keeps walking over the curve.



Point and Cycle attractors

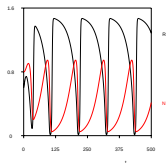
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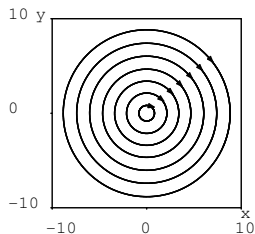
A stable limitcycle is a *closed curve attractor*

- It's a series of (x, y) points, forming closed curve.
- Once it is reached, the system keeps walking over the curve.
- The dynamics of x, y over time are oscillatory.



Dynamics

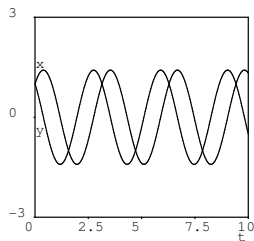
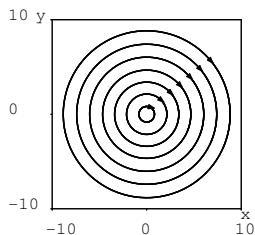
Center point



- A series of closed loops.
- Rotating dynamics in the phase plane.

Dynamics

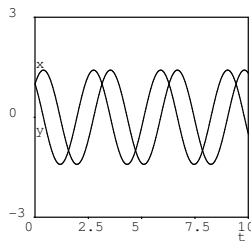
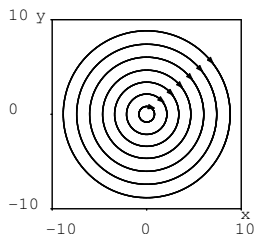
Center point



- A series of closed loops.
- Rotating dynamics in the phase plane.
- Repeated oscillations of the variables.

Dynamics

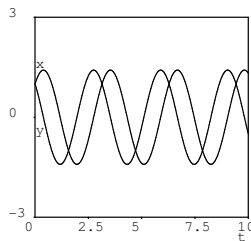
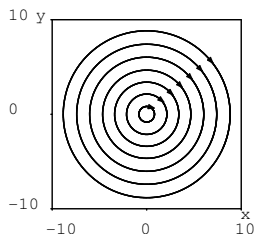
Center point



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- Repeated oscillations of the variables.
- Which loop is followed, depends on initial conditions.

Dynamics

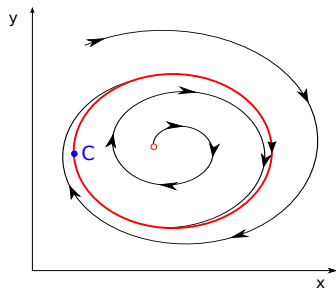
Center point



- A series of closed loops.
- Rotating dynamics in the phase plane.
- Repeated oscillations of the variables.
- Which loop is followed, depends on initial conditions.
- This also determines the amplitude of the oscillations.

Dynamics

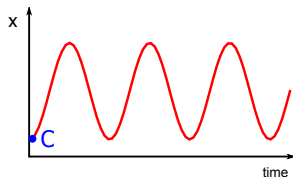
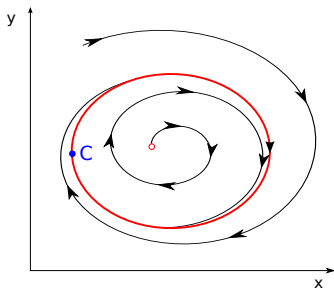
Limit cycle



- A single closed loop.

Dynamics

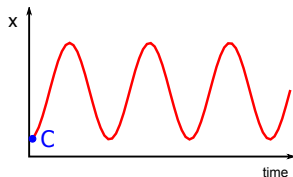
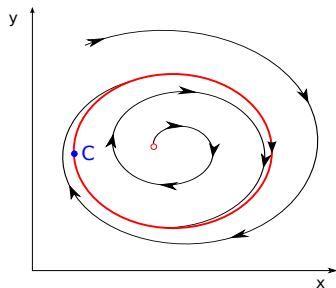
Limit cycle



- A single closed loop.
- So there is only **one single amplitude** of oscillations.

Dynamics

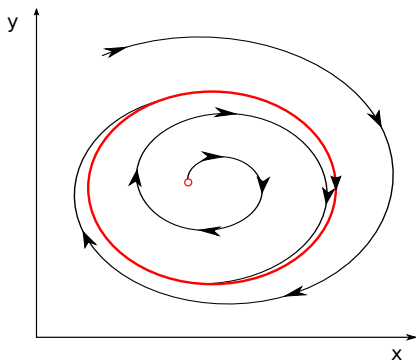
Limit cycle



- A single closed loop.
- So there is only **one single amplitude** of oscillations.
- The stability of the limit cycle determines whether you go to the loop and stay there.

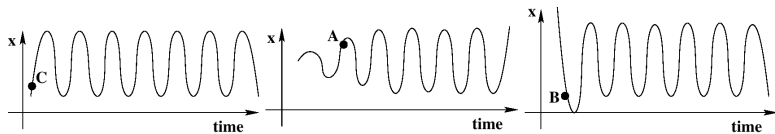
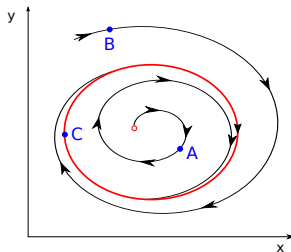
Stable limit cycle

A **stable limit cycle** is an **attractor** consisting of a **closed loop of points**, rather than a single point:



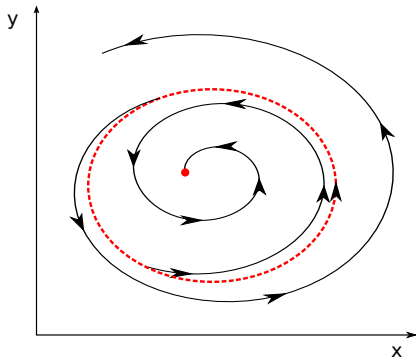
Stable limit cycle

Stable limit cycle: the trajectories converge to the cycle.



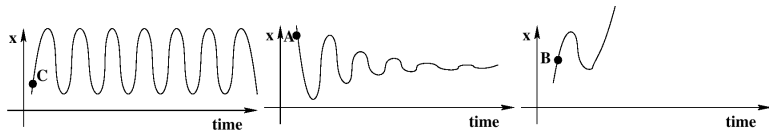
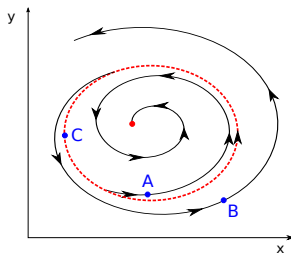
Unstable limit cycle

An **unstable limit cycle** is an **unstable equilibrium**, consisting of a closed loop of points. This loop acts as a **boundary** for the basin of attraction of the attractor that is often within the loop:



Unstable limit cycle

Unstable limit cycle: the trajectories diverge from the cycle.



Why and when do limit cycles occur?

Assume a general non-linear 2D system with a parameter c :

$$\begin{cases} \frac{dx}{dt} = f(x, y, c) \\ \frac{dy}{dt} = g(x, y, c) \end{cases}$$

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Locally, close to the equilibrium, the behaviour changes from a stable to an unstable spiral. However, **globally** nothing changes!

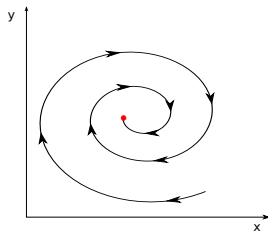
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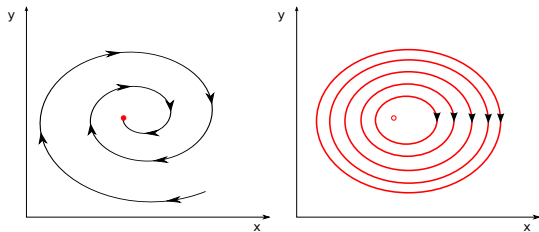
Globally the system converges.



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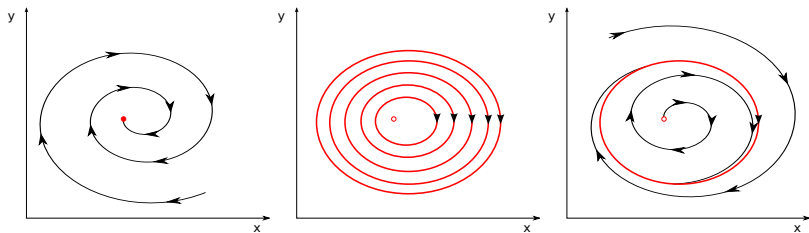
Globally the system converges. But as the real part of the eigenvalues changes from < 0 to > 0 , the spiral becomes locally unstable.



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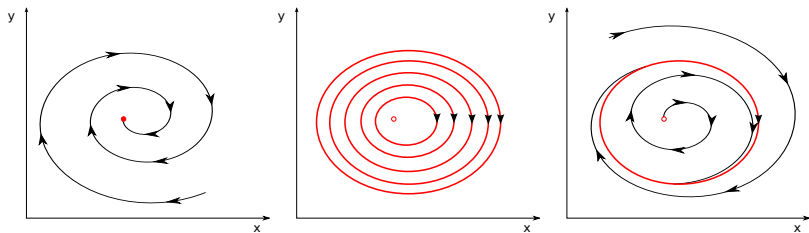
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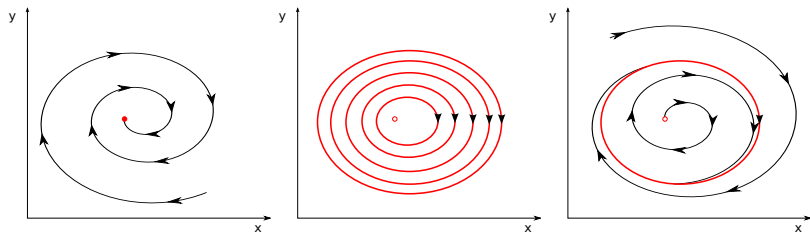


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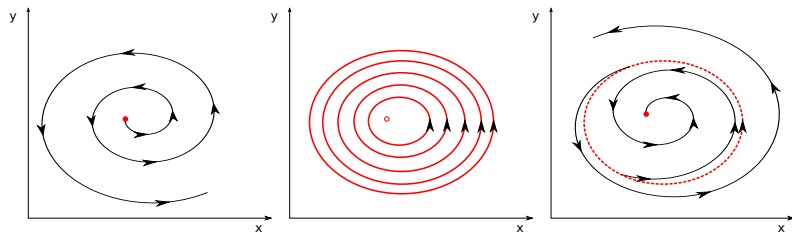


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The local change in stability is called a **Hopf bifurcation**.

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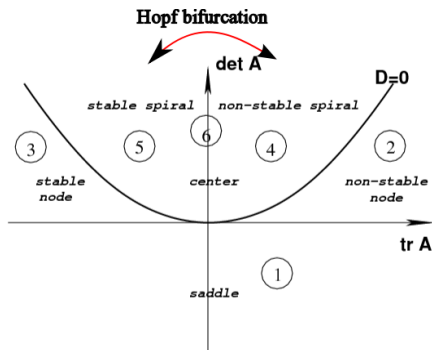
The limit cycle is needed to resolve the conflict between local and global dynamics:

Globally the system diverges. But as the real part of the eigenvalues changes from > 0 to < 0 , the spiral becomes locally stable.



Around the new stable spiral, an unstable limit cycle is needed. The local change in stability is called a **Hopf bifurcation**.

Why and when do limit cycles occur?



2 options:

global dynamics is convergent:

stable spiral $\longleftrightarrow$ unstable spiral
stable limitcycle

global dynamics is divergent:

stable spiral $\longleftrightarrow$ unstable spiral
unstable limitcycle

Example

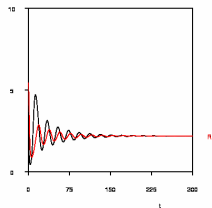
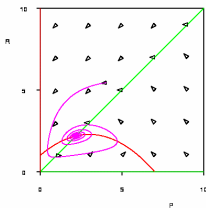
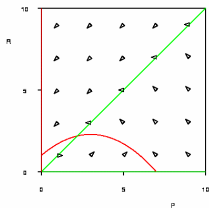
The Holling-Tanner model for predator-prey interactions:

$$\begin{cases} dP/dt = rP(1 - \frac{P}{K}) - \frac{aRP}{d+P} \\ dR/dt = bR(1 - \frac{R}{P}) \end{cases} \quad P > 0; R > 0$$

We fix $a = 1$, $b = 0.2$, $r = 1$, $d = 1$ and vary K .

Example

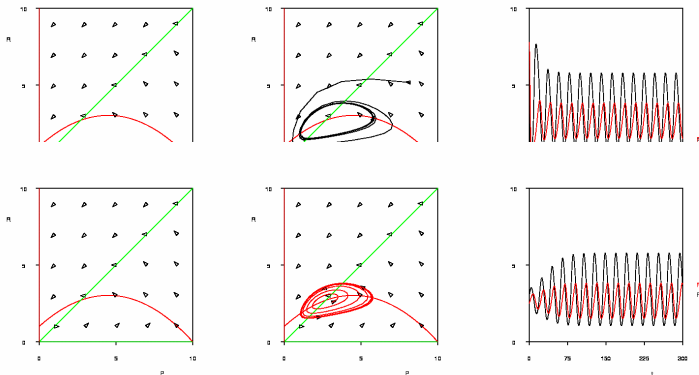
Null-clines and dynamics for $K = 7$:



A stable spiral, with the whole phase plane as a basin of attraction.

Example

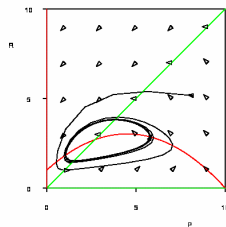
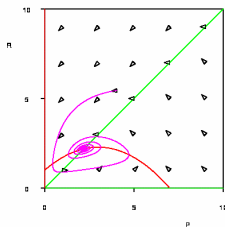
Null-clines and dynamics for $K = 10$:



An unstable spiral and a stable limit cycle, single basin of attraction.

Example

Hopf bifurcation:



Intersection of the nullclines is left of the top in both cases!
Close to top: stable spiral, consistent with the global dynamics.
Further from the top: unstable spiral, with a stable limit cycle.

Limit cycles

Assume a general non-linear 2D system with a parameter c :

$$\begin{cases} \frac{dx}{dt} = f(x, y, c) \\ \frac{dy}{dt} = g(x, y, c) \end{cases}$$

Assume that one of the equilibria has complex eigenvalues:

$$\lambda_{1,2}(c) = \alpha(c) \pm i\beta(c);$$

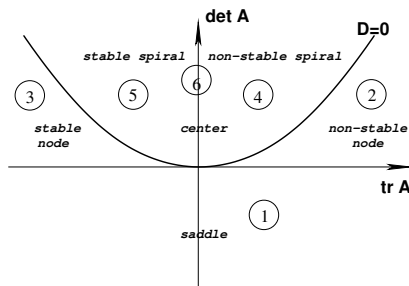
Assume that as c changes, the real part α changes from negative to positive or vice versa.

Locally, close to the equilibrium, the behaviour changes from a stable to an unstable spiral, or vice versa. However, **globally** nothing changes. The limit cycle is needed to resolve the conflict between local and global dynamics.

2D System Analysis

Given a general 2D ODE system:

$$\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$$



- 1 Write $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$, draw the null-clines, draw the vectorfield, find equilibria.
- 2 Look at vectorfield to determine equilibrium types.
- 3 If 2 fails: Use a graphical jacobian to determine the equilibrium type (using the determinant-trace method).
- 4 If 3 fails: Use an analytical jacobian to determine the equilibrium type (using the determinant-trace or the eigenvalue method).